

# Getting Started With Spatial Drawing

<https://gemini.google.com/app/167202fd402ac487>

*User prompt: I would like to get back into drawing. It has been a long time. I would like to be able to draw intricate I ideas. I have found cool concepts like zoom art and mental canvas. I would like to go digital drawing. Where should I start? How should I practice. I have an I pad and Apple Pencil.*

Response: Getting back into drawing is an exciting step, and an iPad paired with an Apple Pencil is arguably the most powerful and intuitive setup you could ask for.

For the specific "zoom art" and "mental canvas" concepts you mentioned, traditional digital art apps (like Procreate or Photoshop) often hit limits. To draw intricate, limitless ideas, you need apps that are built around boundless space.

## Where to Start: The Apps

**Endless Paper** If you want to create "zoom art," this is the industry standard. Endless Paper is a vector-based app, meaning no matter how far you zoom in, your lines will never pixelate or blur. It is essentially one giant, limitless whiteboard where you can nest universes inside of universes.

**Mental Canvas Draw** This app takes your drawings into a completely new dimension. Instead of infinite zoom on a flat 2D plane, Mental Canvas lets you draw 2D strokes in a 3D space. You can create scenes that you actually "fly" through by panning the camera around your layered illustrations, creating a sense of deep architectural space.

**Concepts** If you want to practice intricate ideation before committing to a massive, narrative zoom project, Concepts offers an infinite 2D canvas with excellent toolsets. It's heavily used by concept artists and designers for mind-mapping and sketching without ever running off the edge of the page.

## How to Practice: Your Roadmap Back

Jumping straight into a massive, multi-layered spatial universe after a long break can be frustrating. Because zoom and spatial art rely heavily on scale, transition, and perspective, you need to rebuild your foundational muscle memory first.

### 1 Re-acclimate to the Apple Pencil

Mastering pressure and tilt

Drawing on glass feels much slicker than paper. Spend your first few sessions just making marks. Practice light-to-heavy pressure transitions and tilt the pencil to see how the digital brushes respond. *Tip: If the screen feels too slippery, look into applying a matte, "paper-like" screen protector.*

### 2 Master Basic Shapes in Perspective

The core of spatial drawing

Mental Canvas and zoom art require you to understand how objects look as you move past or into them. Dedicate practice time to drawing spheres, cylinders, and boxes from different

angles. If your basic geometric shapes lack structure, your complex ideas will collapse when viewed from a new spatial angle.

### 3 Plan Scale and Flow

Crucial for infinite zoom

Before drawing a massive zoom piece, sketch a "map." Zoom art works best when there are natural framing devices—like a window, the pupil of an eye, or a doorway—that seamlessly lead the viewer to the next microscopic level. Practice drawing these transition portals.

### 4 Start Small and Iterate

Don't aim for a 100-level zoom art piece on day one. Create a simple 3-level zoom (e.g., a room -> a book on a desk -> a tiny world inside the book). Complete the loop, learn how the app's zooming engine handles the transition, and scale up your ambition on the next project.

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*User prompt: I would like to become versed in technical drawings as well.*

Response: Technical drawing is an entirely different discipline from artistic or spatial drawing. While zoom art is about endless narrative scale, technical drawing is about absolute precision and communicating exactly how a physical object is built.

In mechatronics, technical drawing is the critical bridge between a conceptual idea and a physical, functional part. Whether you are drafting a new gear assembly or designing custom mounts to extrude on your Bambu Lab printer, you need to understand how to communicate dimensions, tolerances, and assembly structures before you ever open FreeCAD or Blender.

## The Two Core Views

Technical drawing relies on standardized ways of projecting 3D objects onto a 2D plane so that anyone, anywhere can manufacture the part without ambiguity.

### 1. Orthographic Projection (Multiview)

This is the workhorse of engineering. It breaks a 3D object down into flat, 2D views—typically the **Top, Front, and Right Side**.

- **The Rule:** You look at the object perfectly straight-on. There is no perspective and no depth.
- **The Purpose:** This is where you place all your precise dimensions, hole diameters, and tolerances. It tells the machinist (or your CAD software) exactly how big the part needs to be.

### 2. Isometric Projection

This is a 3D representation where the object is rotated 45 degrees and tilted 35.26 degrees.

- **The Rule:** Vertical lines stay vertical, but horizontal lines are drawn at 30-degree angles. Unlike a vanishing-point perspective in art, the lines never converge; parallel lines stay parallel forever.

- **The Purpose:** It provides a realistic, immediate understanding of what the assembled part looks like, making the flat orthographic views easier to interpret.

## How to Practice on Your iPad

While professional technical drawings are usually done in CAD software today, sketching them by hand is how you learn the spatial reasoning required to be a good engineer.

### 1 Set Up an Isometric Grid

Use Concepts or Procreate

Don't draw on a blank canvas. Apps like *Concepts* (which is vector-based and great for this) or *Procreate* allow you to turn on drawing guides. Enable an **Isometric Grid**. This forces your lines to snap to the correct 30-degree angles, letting you focus on the shape rather than fighting the perspective.

### 2 Practice the Block-Out Method

Start with primitives

Draw a light bounding box representing the maximum length, width, and height of your object. Then, "carve" away the negative space. If you are drawing a stepped motor mount, draw the whole block first, then slice out the steps.

### 3 Deconstruct Real Objects

Grab something off your desk—a hole punch, a printed 3D part, or a bracket. Try to draw its three orthographic views (Top, Front, Side) on a standard square grid. Then, try to construct the isometric view of that same object using only the flat views you just drew.

### 4 Learn Standard Line Types

Technical drawing is a language. Learn the alphabet:

- **Visible lines:** Thick and solid (outlines of the part).
- **Hidden lines:** Dashed (features that exist but are hidden behind a surface).
- **Center lines:** Long dash, short dash (marks the exact center of holes or symmetrical parts).

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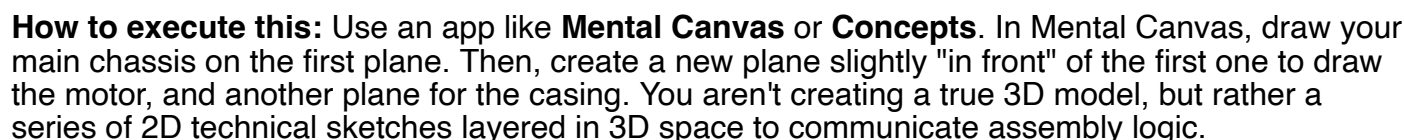
*User prompt: See I would like to combine the two. The infinite canvas 3d idea with technical drawings.*

*Response:* That is exactly how top-tier industrial designers and system architects brainstorm before they ever hit the rigid constraints of traditional desktop software. You are talking about bridging spatial ideation with functional precision.

For complex mechatronics projects, an infinite canvas lets you map an entire system in one place. You can sketch the precise isometric view of a custom 3D-printed mount, pan over to draw its wiring schematic, and zoom out to map the software logic that drives it all.

## The Spatial Architecture Approach

Instead of a flat isometric drawing, you draw the individual components of a mechanical assembly on different 3D planes. When you pan the camera, the assembly visually "pulls apart," showing how the gears, mounts, and casings interlock.



## The Ultimate Bridge: Shapr3D

It is the absolute best bridge between drawing with an Apple Pencil and pure technical engineering. It is a direct-modeling CAD engine built explicitly for the iPad.

- **It feels like drawing:** You sketch 2D profiles using the Apple Pencil just like you would on paper. But because the engine understands geometry, when you drag that sketch, it instantly extrudes into a perfect, mathematically precise 3D solid.
- **The Infinite Workspace:** You can model an entire mechanism in an infinite 3D workspace,

zooming in to adjust a single 2mm tolerance on a gear, and zooming out to see the entire robot chassis.

- **Desktop Handoff:** When you need to transition from these spatial concepts to actual manufacturing—like slicing a part for a Bambu printer—Shapr3D models can be exported as STEP files straight into desktop tools like FreeCAD or Blender for finalization.
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*User prompt: I am inspired by this. It was done in mental canvas. I think it creates to opportunity for the artistry and the detail <https://www.reddit.com/r/Naruto/s/IGwL5UM9G3>*

Response: Those cinematic Mental Canvas videos are incredibly inspiring because they master the art of **parallax**—placing 2D drawings on different 3D planes so that as the camera pushes forward, the foreground elements wipe away to reveal new, intricate worlds behind them.

Combining that level of narrative artistry with technical, mechanical drawing is a brilliant concept. Instead of flying through a village or a character's memories, imagine starting with a beautiful, stylized sketch of a robotic chassis. As the camera pushes forward, it flies *through* the outer casing, revealing the internal mechatronic assemblies, and finally zooming all the way into a microscopic, perfectly dimensioned orthographic blueprint of a single gear.

Here is how you can practice combining artistic spatial storytelling with technical precision using Mental Canvas.

## The Spatial-Technical Workflow

To achieve that seamless fly-through effect without your mechanical drawings turning into a chaotic mess of intersecting lines, you have to build your scene backwards.

### 1 Draft the Core Blueprint

The Micro-Detail

Start on the deepest plane in your Mental Canvas workspace. This is where your pure technical drawing lives. Draw the strict orthographic views (Top, Front, Side) of a specific, intricate part—like a motor mount or a complex joint. Use precise lines, hatching, and dimensions.

### 2 Build the Spatial Assembly

The Exploded View

Create a new drawing plane slightly in front of your blueprint. Here, draw the isometric view of the part you just drafted. Move the camera forward again, create another plane, and draw the components that connect to it. You are essentially drawing an exploded view, but instead of putting all the parts on one piece of paper, you are spacing them out physically in the app's 3D environment.

### 3 Create the 'Portal'

The Transition

To get the cinematic effect from the video you shared, you need a framing device that the

camera flies through to transition from the "art" world to the "technical" world. Create a plane in front of your assembly and draw something like a glowing holographic display, a magnifying glass, or a cracked outer shell.

#### 4 Draw the Establishing Shot

The Artistry

On the very front plane, draw the fully assembled object in a more fluid, artistic style. This is your hero shot. Add shading, environmental context, and stylistic flair.

When you hit play, the camera will start on your artistic hero shot, push through the portal you designed, fly past the floating isometric components, and finally rest on the stark, precise technical blueprint at the back.

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*User prompt: I'd like a drawing practice curriculum. Let's master the basics*

Response: To combine the boundless creativity of spatial art with the rigid precision of engineering, you have to retrain how your arm moves and how your brain parses 3D space.

This 4-week curriculum focuses on draftsmanship—the ability to draw clean, accurate geometry from your imagination without relying entirely on software calculation.

## Week 1: Muscle Memory and Line Quality

Technical drawing requires you to break the habit of drawing from the wrist. The wrist creates arcs; the shoulder creates straight, confident lines.

- **The "Ghosting" Method:** Draw two dots on your iPad. Hover your Apple Pencil over the glass and practice the motion from the shoulder several times before touching down. When you finally make contact, execute the line in one smooth motion.
- **Superimposed Lines:** Draw a straight line using your app's line-snap tool. Then, freehand trace over that exact line 8 times. It will be messy at first. The goal is a confident, single stroke without "wobbling" or fraying at the ends.
- **Target Practice:** Fill a canvas with dots. Connect them randomly with rapid, straight lines. Do not erase if you miss—just analyze why you overshot or curved the line, and try again.

## Week 2: Mastering the Ellipse

An ellipse is just a circle in perspective. It is the absolute foundation of drawing gears, cylinders, shafts, and hardware in 3D space.

- **Ellipses in Planes:** Draw random quadrilaterals in perspective. Bisect them with an X to find the center, then divide them with a cross. Draw an ellipse that touches all four sides perfectly. This teaches you how circular objects distort when rotated away from the camera.
- **The Minor Axis Rule:** The shortest diameter of an ellipse (the minor axis) must always align with the center axis of the cylinder it belongs to. Practice drawing free-floating cylinders, ensuring the circular end-caps follow this mathematical rule. If the ellipse is skewed, the

whole cylinder will look broken.

## Week 3: Isometric Construction

Before flying through an exploded mechatronics assembly, you need to be able to build solid, interlocking forms on a rigid grid.

- **The Carving Exercise:** Turn on the isometric grid guide in your drawing app. Draw a large, light bounding box. Practice "carving" negative space out of it to create stepped motor mounts, angled brackets, and intricate joints. Always start with the maximum volume and cut away.
- **Reverse Engineering:** Take a physical part you've previously modeled in FreeCAD or Blender. Set it on your desk and draw its three flat orthographic views (Top, Front, Side). Then, put the physical part away and try to reconstruct its 3D isometric view using only your flat 2D sketches.

## Week 4: Spatial Deconstruction

This is where the artistic zoom and technical precision merge. You will practice laying out complex systems in 3D space.

- **Multi-Plane Assemblies:** Open Mental Canvas. Draw an outer shell or chassis on the closest plane. Move the camera slightly forward, create a new plane, and draw the internal mechanism (like a gear train, belt drive, or stepper motor) that lives inside that shell.
- **Line Weight Hierarchy:** As your assemblies get complex, they will become visually confusing. Practice assigning strict line weights: use thick, bold lines for the outer silhouettes of parts, and thin, light lines for inner details, hidden edges, and center axes. This makes highly technical drawings instantly readable.

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*User prompt: I wonder if this could be a flutter app that generates practice canvases with the exercise outlined.*

Response: It is completely feasible—in fact, an iPad practice app with procedurally generated drills is an ideal use case for Flutter.

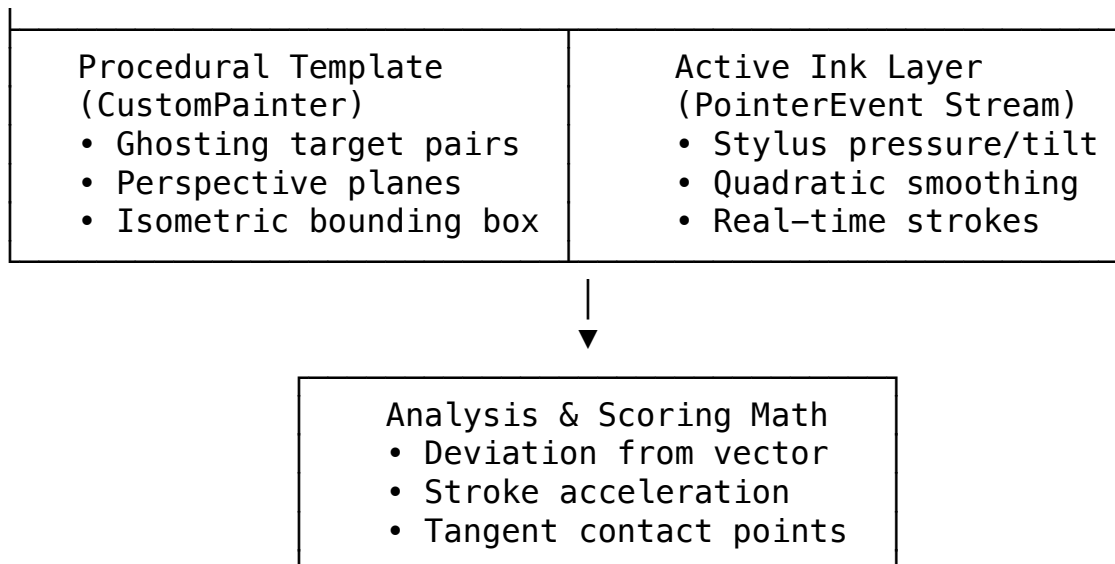
With Impeller as the rendering engine on iOS, Flutter easily hits low-latency 120 Hz ProMotion rendering, which is critical for making an Apple Pencil feel responsive rather than "laggy."

Here is a blueprint of how you could architect an app like this:

### 1. High-Level Architecture

The app essentially splits into three distinct subsystems:

|                                                                           |
|---------------------------------------------------------------------------|
| Interactive View<br>(Custom Matrix4 Transformation for Infinite Pan/Zoom) |
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## 2. Low-Latency Stylus & Stroke Capture

To capture genuine technical drawing input, you need raw hardware events rather than high-level gestures. You can filter for stylus events to isolate pencil input from finger gestures (allowing two-finger pan/zoom while drawing with the tip):

```
Listener(
  onPointerDown: (event) {
    if (event.kind == PointerDeviceKind.stylus) {
      _strokeBuffer.startStroke(
        point: event.localPosition,
        pressure: event.pressure, // 0.0 to 1.0
        tilt: event.tilt,          // Angle relative to surface
      );
    }
  },
  onPointerMove: (event) {
    if (event.kind == PointerDeviceKind.stylus) {
      _strokeBuffer.addPoint(event.localPosition, event.pressure);
    }
  },
  onPointerUp: (event) => _strokeBuffer.commitCurrentStroke(),
  child: CustomPaint(
    painter: CanvasPainter(strokes: _strokeBuffer.strokes),
  ),
);
```

To prevent "jagged" lines on fast strokes, pass the raw points through Catmull-Rom splines or mid-point quadratic Bézier curves before feeding them to a Path.

## 3. Procedural Drill Generators

Instead of loading static background images, you can write mathematical generators that generate

infinite, unique practice canvases on the fly.

## Drill A: The Ghosting Engine

- **The Math:** Pick two points  $P_1 = (x_1, y_1)$  and  $P_2 = (x_2, y_2)$  constrained to a minimum distance.
- **Visual Guide:** Render  $P_1$  (origin) and  $P_2$  (target) as distinct reticles, with a faint, low-opacity baseline connecting them.
- **Analysis Check:** When the user connects the dots, sample points along their stroke and compute the orthogonal distance to the line segment  $\overline{P_1P_2}$ . You can calculate:
  1. **Linear Error:** Average deviation from the true path.
  2. **Velocity Consistency:** Calculate  $\frac{d^2s}{dt^2}$  (acceleration changes) across timestamps. Rapid acceleration drops reveal hesitation and wobble.

## Drill B: Perspective Ellipse Generator

- **The Math:** Generate four points representing a quadrilateral in single-point or two-point perspective.
- **Visual Guide:** Render the quadrilateral with diagonal center lines ( $P_1 \rightarrow P_3, P_2 \rightarrow P_4$ ) to show the perspective center.
- **Analysis Check:** Compute the true inscribed conic section that is tangent to all four quadrilateral edges. Compare the user's closed stroke against the parametric coordinates of the true ellipse to score their accuracy and minor-axis alignment.

## Drill C: Isometric Carve-Outs

- **The Math:** Generate an isometric 3D bounding box (angles locked at  $30^\circ$ ). Subdivide the volume into a  $3 \times 3 \times 3$  voxel grid.
- **Visual Guide:** Randomly "carve" away 1 to 3 corners or steps. Render the resulting edges as faint guidelines.
- **Exercise:** The user must trace the visible outer silhouette with heavy lines, inner steps with medium lines, and hidden internal geometry with dashed lines.

# 4. Canvas State & Memory Management

Because infinite zoom and vector strokes consume memory quickly:

1. **Two-Layer Rendering:** Keep the procedurally generated exercise on a static background layer that only repaints on zoom or new exercise generation. Only repaint the active stroke layer during touch events.
2. **Stroke Serialization:** Store strokes simply as lists of `Offset`, `pressure`, and `timestamp`. This keeps file sizes negligible (a few kilobytes per canvas) and allows you to "replay" the

user's stroke at full speed to analyze where their hand hesitated.

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